

Executive classifier decision guide

State length × question count × GPU utilization × quality

Fixed labels

ModernBERT-large + heads

economics depend on utilization

Runtime labels

GLiClass Modern Large v3

self-hosted flexibility

Managed API

TypeSafe.ai JEV Noul

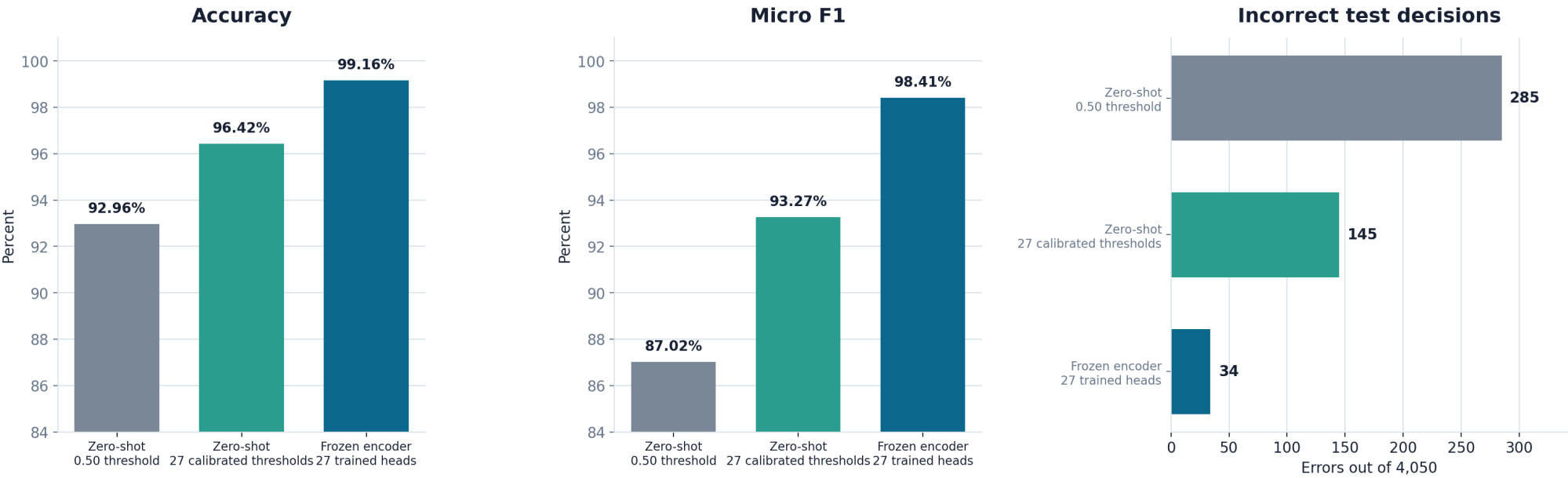
usage-priced; no idle GPU

Mark Austin

Independent synthetic evaluation · September 20, 2026

Threshold calibration captures much of training's gain without changing model weights

ModernBERT-large · same 4,050 locked test decisions · calibration closes 55.8% of the accuracy gap and 54.8% of the F1 gap

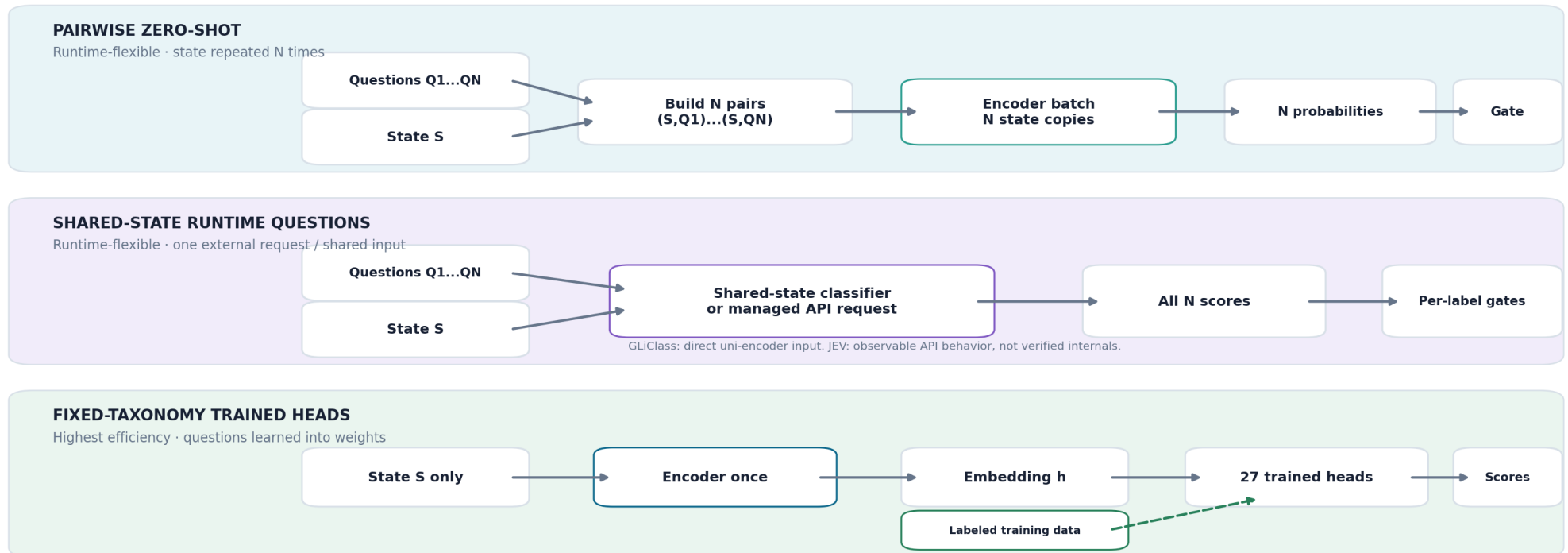


Calibration changes only 27 scalar cutoffs: accuracy rises 3.46 points and errors fall from 285 to 145. Trained heads learn 27 x (1,024 weights + intercept), reaching 34 errors.

The zero-shot architecture still accepts new questions at runtime, but a new question does not inherit a validated label-specific threshold. Trained heads require labeled data and retraining.

Three ways to combine state and classification questions

The efficiency/flexibility trade-off is determined by where question meaning enters the system



Calibration changes gates after scoring; it does not change whether the state is repeated, shared, or encoded once for fixed learned heads.

Fixed taxonomy: which approach has the lowest modeled cost?

Trained ModernBERT heads are eligible · state length × taxonomy size × paid GPU utilization

ModernBERT-large + trained heads TypeSafe.ai JEV Noul

10% paid H100 utilization

10	Trained MB \$0.082	JEV \$0.208	JEV \$0.376	JEV \$0.712	JEV \$1.434
25	Trained MB \$0.082	JEV \$0.268	JEV \$0.436	JEV \$0.772	JEV \$1.554
50	Trained MB \$0.082	Trained MB \$0.326	JEV \$0.535	JEV \$0.871	JEV \$1.753
100	Trained MB \$0.082	Trained MB \$0.326	Trained MB \$0.652	JEV \$1.070	JEV \$2.151

25% paid H100 utilization

10	Trained MB \$0.033	Trained MB \$0.131	Trained MB \$0.261	Trained MB \$0.530	Trained MB \$1.077
25	Trained MB \$0.033	Trained MB \$0.131	Trained MB \$0.261	Trained MB \$0.530	Trained MB \$1.077
50	Trained MB \$0.033	Trained MB \$0.131	Trained MB \$0.261	Trained MB \$0.530	Trained MB \$1.077
100	Trained MB \$0.033	Trained MB \$0.131	Trained MB \$0.261	Trained MB \$0.530	Trained MB \$1.077

50% paid H100 utilization

10	Trained MB \$0.016	Trained MB \$0.065	Trained MB \$0.130	Trained MB \$0.265	Trained MB \$0.539
25	Trained MB \$0.016	Trained MB \$0.065	Trained MB \$0.130	Trained MB \$0.265	Trained MB \$0.539
50	Trained MB \$0.016	Trained MB \$0.065	Trained MB \$0.130	Trained MB \$0.265	Trained MB \$0.539
100	Trained MB \$0.016	Trained MB \$0.065	Trained MB \$0.130	Trained MB \$0.265	Trained MB \$0.539

100% paid H100 utilization

10	Trained MB \$0.008	Trained MB \$0.033	Trained MB \$0.065	Trained MB \$0.133	Trained MB \$0.269
25	Trained MB \$0.008	Trained MB \$0.033	Trained MB \$0.065	Trained MB \$0.133	Trained MB \$0.269
50	Trained MB \$0.008	Trained MB \$0.033	Trained MB \$0.065	Trained MB \$0.133	Trained MB \$0.269
100	Trained MB \$0.008	Trained MB \$0.033	Trained MB \$0.065	Trained MB \$0.133	Trained MB \$0.269

Each cell shows the winner and USD per 1,000 states. ModernBERT-large + trained heads encodes the state once and has negligible incremental head cost, but new or changed questions require labeled data and retraining.

Runtime questions: lowest-cost approach

Trained ModernBERT heads are ineligible · state length × question count × paid GPU utilization

TypeSafe.ai JEV Noul GLiClass Modern Large v3

10% paid H100 utilization

Runtime questions	10	JEV \$0.082	JEV \$0.208	JEV \$0.376	JEV \$0.712	JEV \$1.434
	25	GLiClass \$0.104	JEV \$0.268	JEV \$0.436	JEV \$0.772	JEV \$1.554
	50	GLiClass \$0.126	JEV \$0.367	JEV \$0.535	JEV \$0.871	JEV \$1.753
	100	GLiClass \$0.171	GLiClass \$0.416	JEV \$0.734	JEV \$1.070	JEV \$2.151

25% paid H100 utilization

GLiClass \$0.036	GLiClass \$0.134	GLiClass \$0.265	GLiClass \$0.549	GLiClass \$1.095
GLiClass \$0.042	GLiClass \$0.139	GLiClass \$0.287	GLiClass \$0.565	GLiClass \$1.122
GLiClass \$0.051	GLiClass \$0.148	GLiClass \$0.305	GLiClass \$0.592	GLiClass \$1.167
GLiClass \$0.068	GLiClass \$0.166	GLiClass \$0.341	GLiClass \$0.646	GLiClass \$1.257

50% paid H100 utilization

Runtime questions	10	GLiClass \$0.018	GLiClass \$0.067	GLiClass \$0.132	GLiClass \$0.275	GLiClass \$0.548
	25	GLiClass \$0.021	GLiClass \$0.070	GLiClass \$0.144	GLiClass \$0.283	GLiClass \$0.561
	50	GLiClass \$0.025	GLiClass \$0.074	GLiClass \$0.153	GLiClass \$0.296	GLiClass \$0.583
	100	GLiClass \$0.034	GLiClass \$0.083	GLiClass \$0.171	GLiClass \$0.323	GLiClass \$0.628
		1k	4k	8k	16k	32k

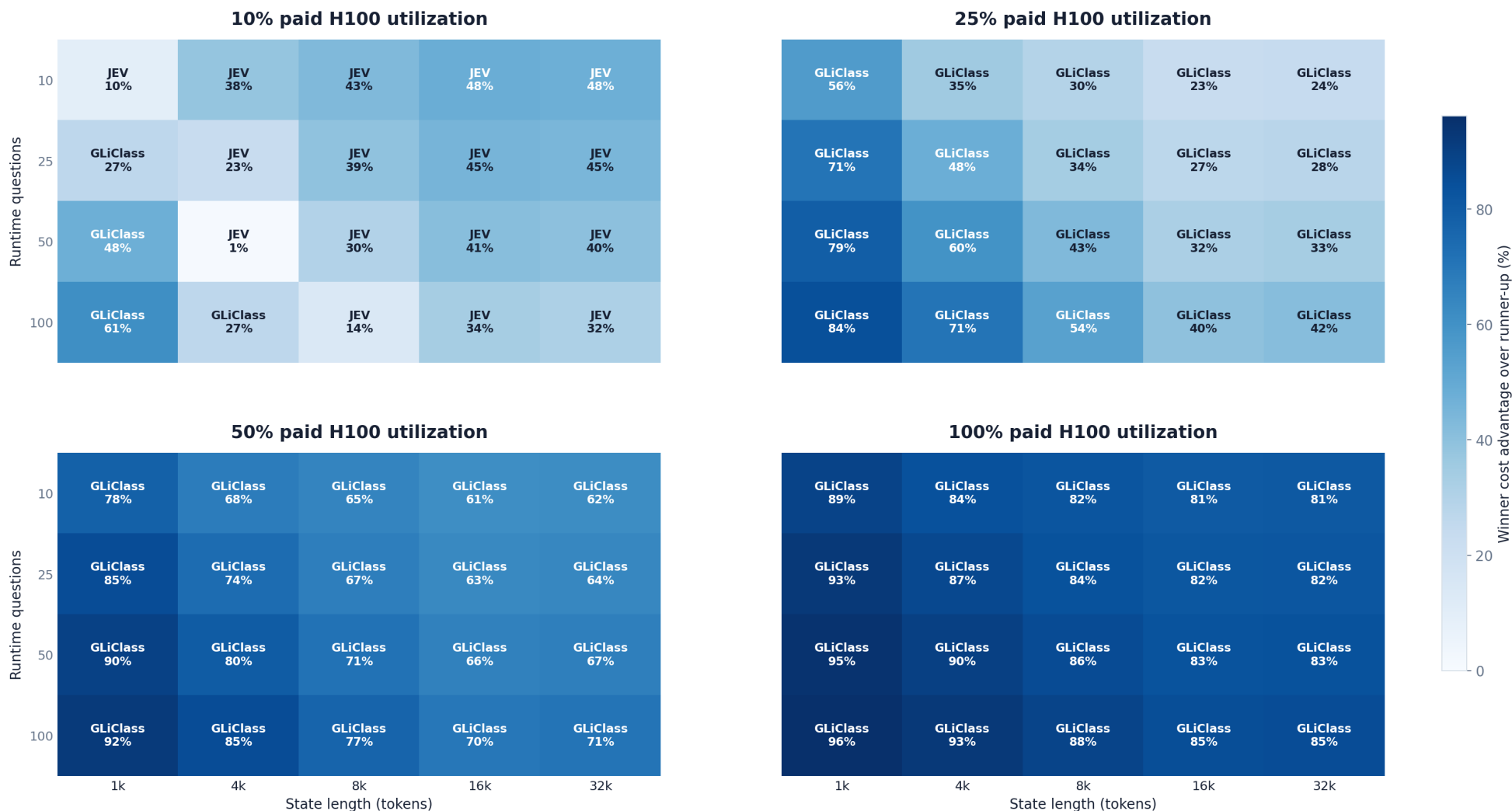
100% paid H100 utilization

GLiClass \$0.009	GLiClass \$0.034	GLiClass \$0.066	GLiClass \$0.137	GLiClass \$0.274
GLiClass \$0.010	GLiClass \$0.035	GLiClass \$0.072	GLiClass \$0.141	GLiClass \$0.280
GLiClass \$0.013	GLiClass \$0.037	GLiClass \$0.076	GLiClass \$0.148	GLiClass \$0.292
GLiClass \$0.017	GLiClass \$0.042	GLiClass \$0.085	GLiClass \$0.162	GLiClass \$0.314
1k	4k	8k	16k	32k

Each cell shows the winner and USD per 1,000 states. Runtime candidates evaluated: GLiClass Modern, TypeSafe.ai JEV Noul, ModernBERT NLI, Luna, and Sol. Use the paired fixed-taxonomy map when trained heads are acceptable.

How decisive is the modeled cost winner?

Runtime questions only; trained heads are ineligible · high percentages are robust, low percentages indicate a close decision



Advantage = (runner-up cost - winner cost) / runner-up cost. Operational simplicity, data residency, latency, support, and quality may outweigh a small modeled cost advantage.

Executive decision map: economics versus runtime flexibility

Reference workload: 6,000 state tokens × 25 questions · self-hosted cost at 100% H100 utilization

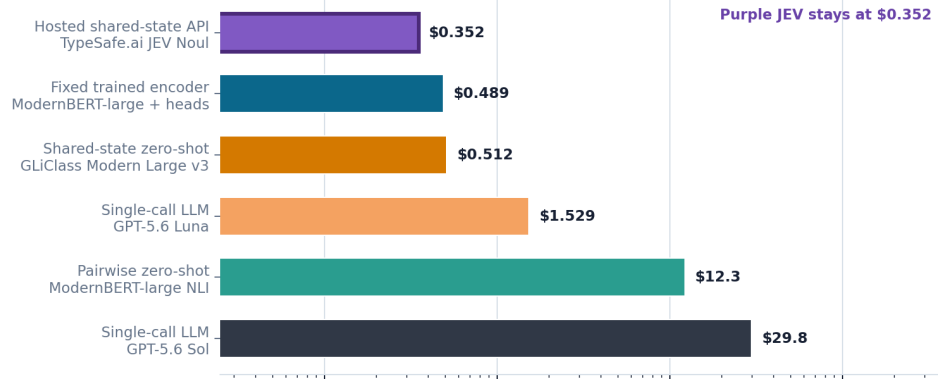


The \$1/1,000 boundary is an executive guide, not a universal purchasing threshold. Flexible systems accept new questions without retraining; trained heads do not.

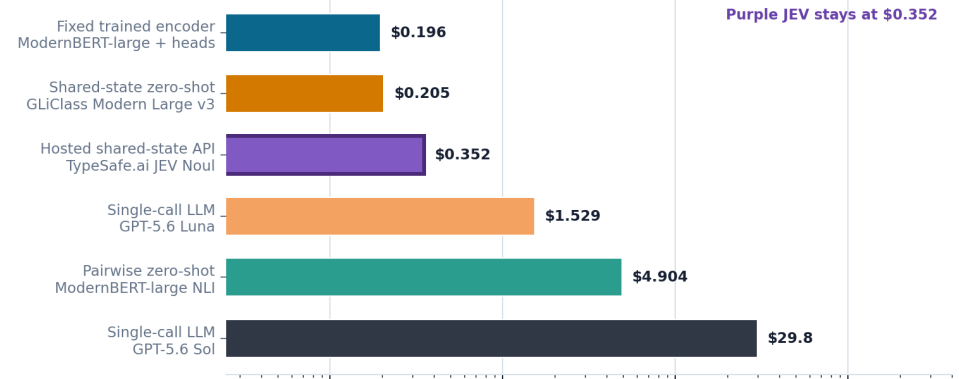
GPU utilization determines self-hosted encoder economics

Fixed workload: 6k state tokens × 25 questions · NVIDIA H100 at USD 5/hour · lower is better

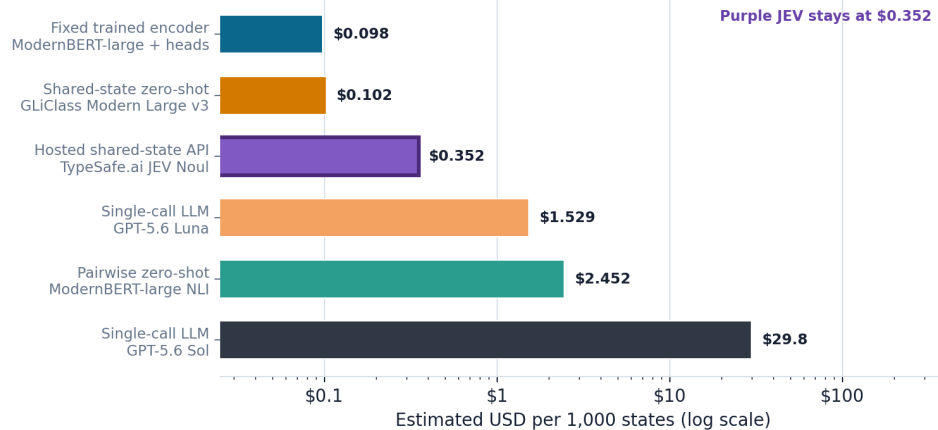
10% paid H100 utilization



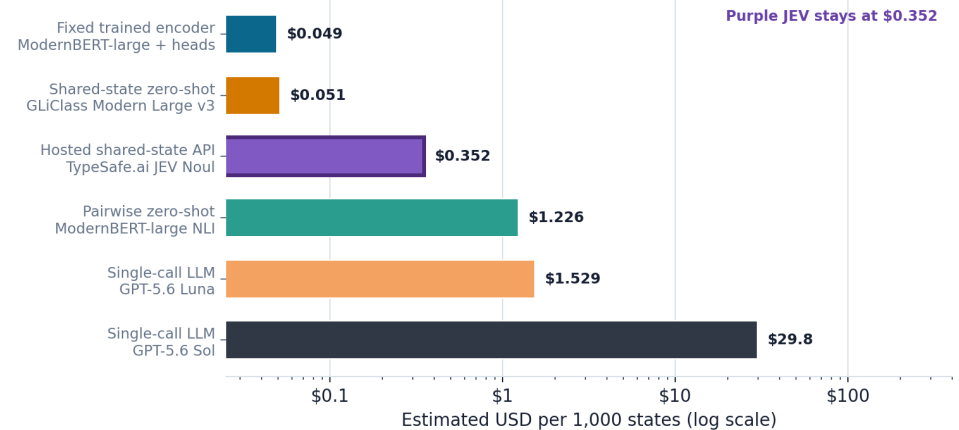
25% paid H100 utilization



50% paid H100 utilization



100% paid H100 utilization



Self-hosted encoder cost is divided by paid GPU utilization; hosted JEV/Luna/Sol prices remain usage-based. In this scenario, JEV crosses trained ModernBERT near 13.9% utilization and GLiClass Modern near 14.6%. Hidden LLM reasoning tokens are excluded.

Operating-model scorecard

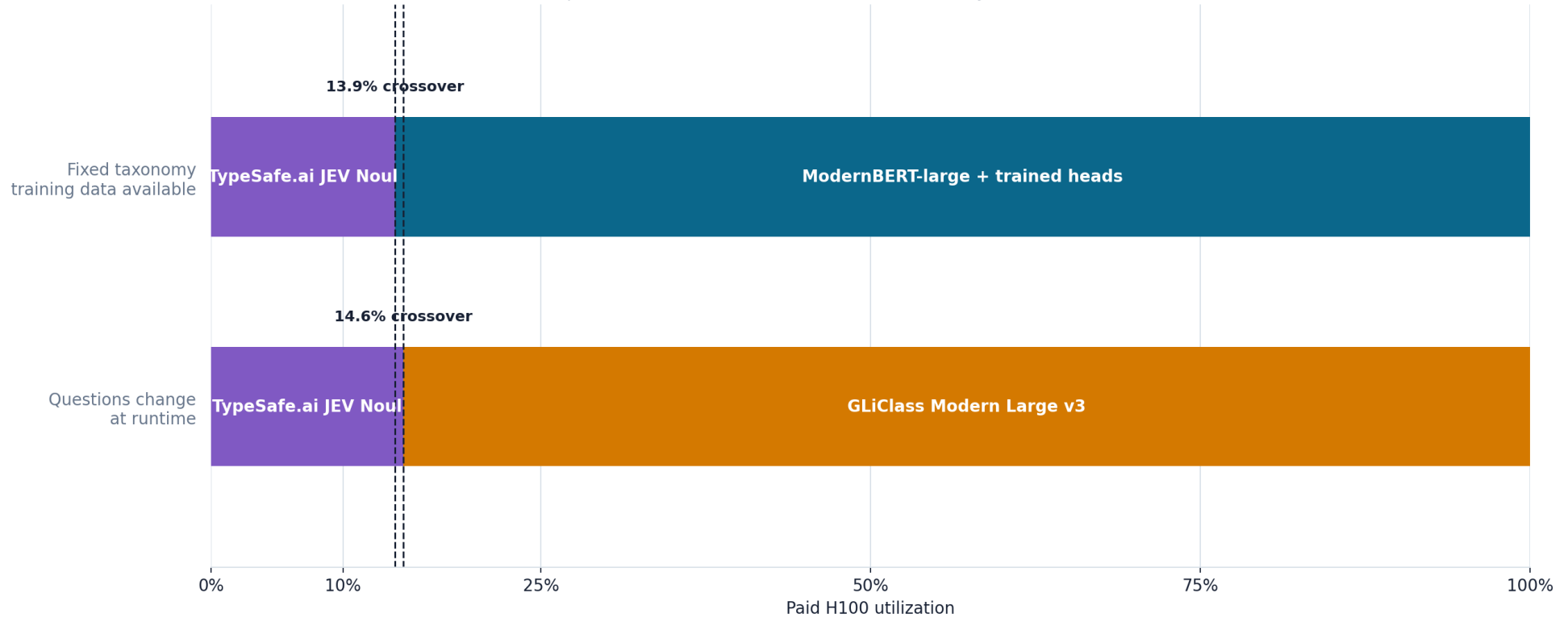
Named examples from this benchmark · choose the operating model before choosing the model

Operating pattern	Example implementation	Accuracy	Runtime questions	Deployment	Executive takeaway
Fixed trained encoder	ModernBERT-large + trained logistic heads	99.16%	No	Self-hosted	Default economic path when taxonomy is stable
Shared-state zero-shot	GLiClass Modern Large v3	97.90%	Yes	Self-hosted	Flexible winner above ~14.6% utilization at 6k × 25
Hosted shared-state API	TypeSafe.ai JEV Noul	99.90%	Yes	Managed API	Usage-priced; no customer-owned idle GPU
Pairwise zero-shot	ModernBERT-large zero-shot NLI	96.42%	Yes	Self-hosted	Repeats state for every question
Single-call LLM	GPT-5.6 Luna / GPT-5.6 Sol	99.75% / 99.93%	Yes	Managed API	Premium option for broader semantic reasoning

Accuracy is measured on the locked 150-transcript synthetic test. The 14.6% utilization crossover is specific to the 6k-state-token × 25-question, USD 5/H100-hour scenario.

Which operating model wins as GPU utilization changes?

Reference workload: 6k state tokens × 25 questions · H100 at USD 5/hour · modeled cost only



Below the crossover, usage-priced JEU is cheaper than carrying idle H100 capacity. Above it, self-hosted encoders win on modeled marginal cost. Operational requirements and measured quality remain separate gates.

What should an executive choose?

Select the operating model before selecting the model

Known, stable taxonomy

ModernBERT-large + trained heads

Best modeled economics with 99.16% test accuracy when utilization is sustained. New questions require labeled data and retraining.

Runtime labels, self-hosted

GLiClass Modern Large v3

Best open zero-shot trade-off tested. In the reference workload it becomes cheaper than JEV above about 14.6% paid GPU utilization.

Runtime questions, managed service

TypeSafe.ai JEV Noul

99.90% calibrated accuracy and no customer-owned idle GPU cost. It leads the reference scenario below roughly 14% utilization.

Broad reasoning or maximum test score

Luna or Sol

Near-ceiling quality, but use when semantic breadth justifies the premium. Sol's advantage over calibrated JEV was one decision in 4,050.

Executive guardrails

- Quality results come from a synthetic, explicit-evidence test and are not production guarantees.
- Cost curves are modeled from token shapes, published/API rates, and H100 throughput proxies.
- Validate with real states, real questions, target hardware, and operational error costs before procurement.